CHE 321 HW 3 Veronica Gebbia

P4-1$_A$ (a) **Example 4-1.** Would the example be correct if water were considered an inert? Explain.
(b) **Example 4-2.** How would the answer change if the initial concentration of glyceryl stearate were 3 mol/dm^3? Rework Example 4-2 correctly using the information given in the problem statement.
(c) **Example 4-3.** Under what conditions will the concentration of the inert nitrogen be constant? Plot Equation (E4-5.2) in terms of $(1/-r_A)$ as a function of X up to value of $X = 0.99$. What did you find?
(e) **Example 4-4.** Why is the equilibrium conversion lower for the batch system than the flow system? Will this always be the case for constant-volume batch systems? For the case in which the total concentration C_{T0} is to remain constant as the inerts are varied, plot the equilibrium conversion as a function of the mole fraction of inerts for both a PFR and a constant-volume batch reactor. The pressure and temperature are constant at 2 atm and 340 K. Only N_2O_4 and inert I are to be fed. Go to the Living Example Problems and load Wolfram. (1) What values of K_C, C_{A0}, and ε cause X_{ef} to be the farthest away from X_{eb}? (2) At what value C_{A0} will X_{eb} and X_{ef} be closest together?
(d) **Example 4-5.** **(a)** Using the molar flow rate of A of 3 mol/minute and Figure E4-5.1, calculate the PFR volume necessary for 40% conversion. **(b)** Next consider the entering flow rate of SO_2 is 1,000 mol/h. Plot $\left(F_{A0}/-r_A'\right)$ as a function of X to determine the PFR catalyst weight to achieve (1) 30% conversion, (2) 40% conversion, and (3) 99% of the equilibrium conversion, i.e., $X = 0.99\ X_e$.

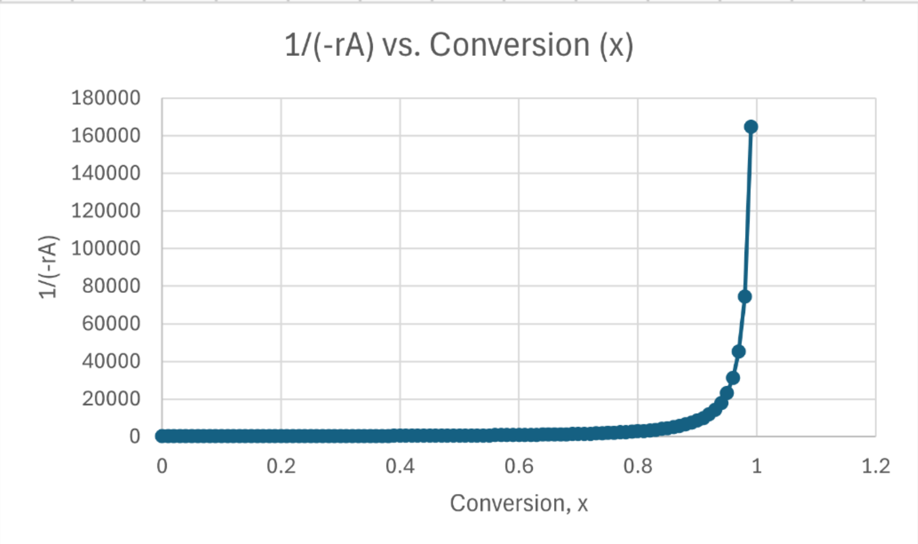
Part (a): H_2O as inert → l-phase batch ($H_2O = w$)
This example is for a liquid-phase batch system which means $v = v_0$. This is due to the reaction happening in the liquid phase. Since H_2O is now inert then the concentration is constant ($C_w = C_{w0}$) and this is just added onto the stoich relations set. With water as inert, it only changes the initial composition which is the θ's and C_{A0} but it doesn't alter $C_j = h_j(x)$. This means everything else is still correct.

Part (b): if $B = 3\ \text{mol/dm}^3$
for this feed: $\theta_B = 3/10 = 0.30$
limiting $= B$ since $C_B > 0 \rightarrow \theta_B - x/3 \geq 0$ so then
$x = 3\theta_B$ and $x = 0.90$ (max convert)
rework: $x = 0.20$
$C_B = 10(0.30 - 0.20/3) = 2.33\ \text{mol/dm}^3$
$C_D = 10(0.20 \div 3) = 0.67\ \text{mol/dm}^3$
$x = 0.90$: all consumed
$C_B = 10(0.30 - 0.90/3) = 0\ \text{mol/dm}^3$
$C_D = 10(0.90 \div 3) = 3\ \text{mol/dm}^3$

When glyceryl stearate concentration increased to $3\ mol/dm^3$ then the stoich ratio became $\theta_B = C_{B0}/C_{A0} \rightarrow = 0.3$. This makes C_B the limiting reactant with a max conversion of 90% for NaOH. At 20% conversion, C_B and C_D have reasonable concentration. Yet, at 90% conversion the B is completely consumed and C_D becomes the $3\ mol/dm^3$. Therefore, 90% is the max because any more would cause a negative C_B and that's impossible. This shows why the limiting reactant must be used since it wasn't possible otherwise.

Part (c):

1) under what conditions will inert N be const.
all the denominators for C_A, C_B, and $C_N = 1+\varepsilon x$
and $\varepsilon = y_{A0}\delta \rightarrow \delta = d/a + c/a - b/a - 1$
for C_N to be constant then $\varepsilon = 0$. This can occur when $\delta = 0$. other requirements are that total molar flow is constant and the inert concentration must be $C_N = C_{N0}$.

2) 

What I saw $\rightarrow$ Levenspiel

The plot showed the slow rise of $1/(-r_A)$ vs. x. The data diverges as x approaches 1. This is very useful to find almost complete conversion yet it may not be realistic.

Part (e):

1) The equilibrium conversion is lower for batch:

$A \rightleftharpoons 2B \rightarrow \delta = d/a + c/a - b/a - 1 \rightarrow +1 \quad \varepsilon = y_{A0}\delta = y_{A0} > 0$

at const. V: $C_A = C_{A0}(1-x)$, $C_B = C_{A0}(2x)$

$K_C = C_B^2/C_A \rightarrow K_C = \dfrac{4C_{A0}x^2}{1-x} \rightarrow (4C_{A0})x^2 + K_C x - K_C = 0$

batch ↑

$C_A = \dfrac{C_{A0}(1-x)}{1+\varepsilon x}$, $C_B = \dfrac{C_{A0}(2x)}{1+\varepsilon x}$ which makes →

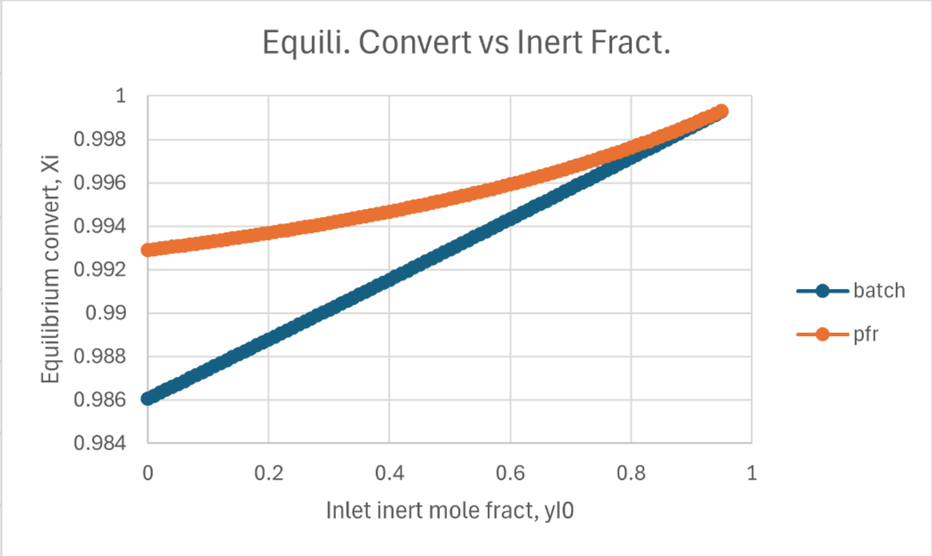
$K_C = \dfrac{4C_{A0}x^2}{(1+\varepsilon x)(1-x)} \rightarrow (4C_{A0} + K_C\varepsilon)x^2 + K_C(1-\varepsilon)x - K_C = 0$

flow ↑

compare the two: $X_{i,flow} > X_{i,batch}$ if $\delta > 0$

b) The case of constant-volume batch systems may not be accurate. This is due to the net mole change of a reaction which helps determine whether batch will give a higher or lower equilibrium conversion versus a flow system. For $A \rightleftharpoons 2B$, $\delta = +1$ & $\varepsilon > 0$ and the gas-flow equilibrium includes the dilution factor of $1+\varepsilon x$. This factor will cause a push toward products compared to a batch vessel that is constant volume and rigid. Therefore, $X_{i,flow} > X_{i,batch}$. Yet, if $\delta < 0$ then the opposite occurs then the inequality flips and if $\delta = 0$ then $X_{i,flow} = X_{i,batch}$.

Plot: →



i) values of K_c, C_{A0}, ε cause X_{ef} to be farthest away from X_{eb}?

$K_c \approx 0.338$ mol/L ; $C_{A0} = 0.071686$ mol/L ; $\varepsilon = 1$

ii) value of C_{A0} = X_{eb}, X_{ef} closest together

$X_{eb} = 0.646$, $X_{ef} = 0.736$ and gap becomes smaller as C_{A0} approaches 0.

Part d:

a: $C_{T0} = P/RT \rightarrow 2/0.0821 \cdot 340 = 0.07165$ mol/L

$k_r = k_f / K_c \rightarrow 0.5/0.1 = 5$ L/mol·min

$C_A = C_{A0}(1-x)/1+x \quad C_B = 2C_{A0}x/1+x$

$-r_A = k_f C_A - k_r C_B^2 = k_f\left(C_{A0}(1-x)/1+x\right) - k_r\left(2C_{A0}x/1+x\right)^2$

PFR: $V = F_{A0} \cdot \int_0^x dx/-r_A$

at $x = 0.40 \rightarrow V = 69.1\text{ L} = 69.1\text{ dm}^3$

b) $-r'_{SO_2}(x) =$

$$9.7\left[\frac{8.3\left(\frac{1-x}{1-0.14x}\right)\sqrt{\frac{1.08-x}{2(1-0.14x)}}-0.0044\left(\frac{x}{1-0.14x}\right)}{\left(1+\sqrt{\frac{79(1.08-x)}{1-0.14x}}+\frac{174(1-x)}{1-0.14x}\right)^2}\right]$$

$-r' = 0$ so $x_e \approx 0.998$

$$W = \int_0^x \frac{F_{AO}}{-r'_{SO_2}(x)}\,dx \quad \text{at } x = 0.3, 0.4, 0.99x_e = 0.988$$

$x = 0.3 : W = 1.61 \times 10^5$

$x = 0.4 : W = 2.1 \times 10^5$

$x = 0.99x_e : W = 4.03 \times 10^5$

#2 P4-11c

Set-up for ease:

$\delta = (d/a + c/a - b/a - 1) \rightarrow = 6$; $y_{A0} = 0.5$

$\varepsilon = y_{A0}\delta = 3$

$N_T/N_{T0} \rightarrow 1 + \varepsilon x \rightarrow = 1 + 3x$

volume: $v = v_0 (N_T/N_{T0})(P_0/P)(T/T_0)$ $T = T_0$

$V = 0.1P \rightarrow P/P_0 = v/v_0$

$v = v_0 (N_T/N_{T0})(v_0/v) \rightarrow$ * $v = v_0\sqrt{N_T/N_{T0}} = v_0\sqrt{1+3x}$

concentration & convert

$N_A = N_{A0}(1-x)$; $N_B = N_{B0} - N_{A0}X = N_{A0}(1-x)$

plug into : $C_A = C_B = \dfrac{N_{A0}(1-x)}{v_0\sqrt{1+3x}} = C_{A0}\left(\dfrac{(1-x)}{\sqrt{1+3x}}\right)$ *

$t=0$: $v_0 = 0.1P_0 \rightarrow P_0 = 1.5\,atm$

$C_{T0} = P_0/RT \rightarrow 1.5/0.73 \cdot 600 = 0.003425\ lbmol/ft^3$

$C_{A0} = C_{B0} = \frac{1}{2}C_{T0} \rightarrow = 0.001712\ lbmol/ft^3$

a) rate law: $-r_A = k_1 C_A^2 C_B \rightarrow$

$-r_A = k_1(C_{A0})^3\left(\dfrac{(1-x)^3}{(1+3x)^{3/2}}\right)$ and $k = 1$ $\rightarrow$

$k_1(C_{A0})^3 = (0.00172)^3 = 5.08 \cdot 10^{-9}\ lbmol/ft^3 s$

$$-r_A(x) = 5.08 \cdot 10^{-9}\left(\dfrac{(1-x)^3}{(1+3x)^{3/2}}\right)\ lbmol/ft^3 s$$

b) $v = 0.2\,ft^3$

$0.2 = 0.15\sqrt{1+3x} \rightarrow (\sqrt{1+3x} = 1.333)^2 \rightarrow 1+3x = 1.778$

$x = 0.2593$

$(1-x)^3 = (1-0.2593)^3 = 0.40644$

$(1+3x)^{3/2} = (1+3(0.2593))^{3/2} = 2.370$

$$-r_A = 5.08 \times 10^{-9} \cdot \left(0.40644/2.370\right) = 8.726 \times 10^{-10}\ \mathrm{lbmol/ft^3 s}$$